Conventions and transformations of ray launching angles

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There are several definitions for the ray launching angles used in ECCD. The most common definitions are listed here.

- for the vertical angle :
  - α $[-\pi;\pi]$: angle between the poloidal projection of the ray and the horizontal plane. Refered to as the poloidal angle in [pra08].
  - $\tilde{\alpha}$ $[-\pi/2;\pi/2]$: angle between the ray and the horizontal plane.
  - θ $[0;\pi]$: angle between the ray and the Z axis [LUKE]. Refered to as the polar angle in [pra08].
- for the horizontal angle :
  - β $[-\pi/2;\pi/2]$: angle between the ray and the poloidal plane. Refered to as the toroidal angle in [pra08].
  - $\tilde{\beta}$ $[-\pi;\pi]$: angle between the horizontal projection of the ray and the poloidal plane.
  - ϕ $[0;2\pi]$: angle between the ray and the R axis [LUKE]. Refered to as the azimuthal angle in [pra08].

These angles are reported on Fig. 1. Note that these definitions are valid both in the case of low-field side ($|\alpha| < \pi/2$, $\left|\tilde{\beta}\right| < \pi/2$) and high-field side ($|\alpha| > \pi/2$, $\left|\tilde{\beta}\right| > \pi/2$) launching. Note that the sign convention is that the angle α, β, $\tilde{\alpha}$ and $\tilde{\beta}$ are positive as indicated by arrows on the figure. The convention for β and $\tilde{\beta}$ depends upon the direction of the toroidal angle. The angles θ and ϕ are defined uniquely with respect to R and Z.

The angles θ and ϕ are similar to the angles $\tilde{\alpha}$ and $\tilde{\beta}$. They are simply related through

$$\boxed{\begin{aligned}\theta &= \frac{\pi}{2} - \tilde{\alpha} \\ \phi &= \pi + \tilde{\beta}\end{aligned}} \tag{1}$$

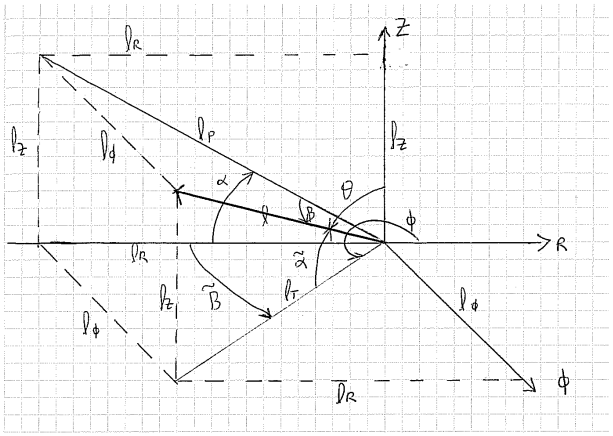


Figure 1: geometry of poloidal and toroidal angles

The relations between α, β, $\tilde{\alpha}$ and $\tilde{\beta}$ are expressed by four geometrical relations

$$\sin\beta = \cos\tilde{\alpha}\sin\tilde{\beta} \quad (2)$$
$$\tan\beta = \cos\alpha\tan\tilde{\beta} \quad (3)$$
$$\sin\tilde{\alpha} = \cos\beta\sin\alpha \quad (4)$$
$$\tan\tilde{\alpha} = \cos\tilde{\beta}\tan\alpha \quad (5)$$

The transformation from (α,β) to $(\tilde{\alpha},\tilde{\beta})$ coordinates is derived from (4,3)_

$$\boxed{\begin{array}{l}\tilde{\alpha} = \arcsin\left(\cos\beta\sin\alpha\right) \\ \tilde{\beta} = \arctan\left(\dfrac{\tan\beta}{\cos\alpha}\right) + \pi\text{sign}(\beta)\left[|\alpha| > \dfrac{\pi}{2}\right]\end{array}} \quad (6)$$

such that the transformation from (α,β) to $(\tilde{\alpha},\tilde{\beta})$ coordinates is (1,6)

$$\boxed{\theta = \arccos\left(\cos\beta\sin\alpha\right)} \quad (7)$$
$$\phi = \pi + \arctan\left(\frac{\tan\beta}{\cos\alpha}\right) + \pi\text{sign}(\beta)\left[|\alpha| > \frac{\pi}{2}\right] \quad (8)$$

Note that instead of (**??**) we can use (2,1)

$$\boxed{\phi = \pi + \arcsin\left(\frac{\sin\beta}{\sin\theta}\right) + \pi\text{sign}(\beta)\left[|\alpha| > \frac{\pi}{2}\right]} \quad (9)$$

which avoids the numerical errors that ay arise from (**??**). The expression (**??**) and (9) were derived in [pra08] for the case of low-field side launching only.

The inverse relation could also be useful. We can transform from $(\tilde{\alpha},\tilde{\beta})$ to (α,β) according to the relations

$$\alpha = \arctan\left(\frac{\tan\tilde{\alpha}}{\cos\tilde{\beta}}\right) + \pi\text{sign}(\tilde{\alpha})\left[\left|\tilde{\beta}\right| > \frac{\pi}{2}\right]$$
$$\boxed{\beta = \arcsin\left(\cos\tilde{\alpha}\sin\tilde{\beta}\right)}$$

Again, we may use instead

$$\boxed{\alpha = \arcsin\left(\frac{\sin\tilde{\alpha}}{\cos\beta}\right) + \pi\text{sign}(\tilde{\alpha})\left[\left|\tilde{\beta}\right| > \frac{\pi}{2}\right]} \quad (10)$$

In addition, it is possible to derive the following cross-transformations : with prior knowledge of LFS or HFS launching, we obtain $(\tilde{\alpha},\beta)$ to $(\alpha,\tilde{\beta})$ by using (10,2)

$$\alpha = \arcsin\left(\frac{\sin\tilde{\alpha}}{\cos\beta}\right) + \pi\text{sign}(\tilde{\alpha})\,[\text{HFS}]$$
$$\tilde{\beta} = \arcsin\left(\frac{\sin\beta}{\cos\tilde{\alpha}}\right) + \pi\text{sign}(\beta)\,[\text{HFS}]$$

We can also derive $(\alpha,\tilde{\beta})$ to $(\tilde{\alpha},\beta)$ from (5)

$$\tilde{\alpha} = \arctan\left(\cos\tilde{\beta}\tan\alpha\right)$$
$$\beta = \arctan\left(\cos\alpha\tan\tilde{\beta}\right)$$

If the launching point is in free space the index of refraction $\mathbf{n}$ is parallel to the ray direction and $\|\mathbf{n}\| = 1$. The parallel index of refraction is

$$n_{\|} \equiv \mathbf{n}\cdot\hat{\mathbf{b}} = \frac{n_R\cdot B_R + n_Z\cdot B_Z + n_\phi\cdot B_\phi}{\|\mathbf{B}\|}$$

where we have the relations

$$n_R = -\cos\alpha\cos\beta = -\cos\tilde{\beta}\cos\tilde{\alpha} = \cos\phi\sin\theta$$
$$n_Z = \sin\alpha\cos\beta \qquad = \sin\tilde{\alpha} = \cos\theta$$
$$n_\phi = \sin\beta \qquad = \sin\tilde{\beta}\cos = -\sin\phi\sin\theta$$